



Data Visualization

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Activity 1: Gender Distribution

Scenario: Student Affairs wants to understand the gender composition of students.



Business Question:
What is the gender distribution of students?



Recommended Visualization: Donut Chart / Bar Chart



Your Task: Create the visualization and write two insights.



Activity 2: School Type

Scenario: Management wants to know which school type contributes the most students.



Business Question:
Which school type has the most students?



Recommended Visualization:
Horizontal Bar Chart



Your Task: Visualize the data and explain the findings.



Activity 3: Mathematics Scores

Scenario: The Mathematics Department wants to understand student performance.



Business Question:
What is the distribution of Mathematics scores?



Recommended Visualization:
Histogram



Your Task: Describe the overall distribution.



Activity 4: Final Grades

Scenario: The Vice-Chancellor needs an overview of student performance.



Business Question: Which final grade occurs most frequently?



Recommended Visualization: Bar Chart



Your Task: Identify the most common grade.



Activity 5: Subject Performance

Scenario: Academic Planning wants to compare subjects.



Business Question:
Which subject has the highest average score?



Recommended Visualization:
Column Chart



Your Task:
Recommend where intervention is needed.



Activity 6: Study Hours

Scenario: Student Development wants evidence on study habits.



Business Question:
Do students who study more perform better?



Recommended Visualization:
Scatter Plot



Your Task: Explain the relationship.



Activity 7: Attendance

Scenario: The university is reviewing its attendance policy.



Business Question:
Does attendance influence performance?



Recommended Visualization: Scatter Plot



Your Task:
Recommend whether attendance should be encouraged.



Activity 8: Internet Access

Scenario:
Management is considering expanding digital support.



Business Question:
Do students with internet access perform better?



Your Task: Provide evidence and recommendations.



Recommended Visualization: Box Plot



Activity 9: Parent Education

Scenario:
Researchers are
studying family
background.



Business Question:
Does parent
education influence
performance?



Your Task: Interpret
the differences.



**Recommended
Visualization:** Box
Plot



Activity 10: Study Method

Scenario: The Teaching and Learning Centre wants to recommend study techniques.



Business Question: Which study method performs best?



Recommended Visualization: Bar Chart



Your Task: Identify the best-performing method.



Activity 11: Age Distribution

Scenario:
Management wants
to understand the
student age profile.



**Business
Question:** How are
students distributed
by age?



Your Task: Describe
the age distribution.



**Recommended
Visualization:**
Histogram



Activity 12: Mathematics Outliers

Scenario: Academic advisors want to identify unusual results.



Business Question: Are there outliers in Mathematics scores?



Recommended Visualization: Box Plot



Your Task: Discuss possible reasons for outliers.



Activity 13: Correlation

Scenario: The Institutional Research Office wants to identify important relationships.



Business Question: Which variables are related to overall performance?



Recommended Visualization: Correlation Heatmap



Your Task: Identify the strongest relationships.



Activity 14: Extra Activities

Scenario: Student Affairs wants to measure student engagement.



Business Question:
How many students participate in extracurricular activities?



Recommended Visualization: Donut Chart



Your Task: Interpret participation levels.



Activity 15: Dashboard Challenge

Scenario: You are presenting to the Vice-Chancellor.



Business Question: What are the five most important insights?



Your Task: Build a dashboard and provide recommendations.



Recommended Visualization:
Dashboard



Student Performance Executive Dashboard

06 June 2026

All Students

DIRISA Datathon 2026 | Data Visualization Practical



Overview



Performance



Engagement



Learning



Reports



Settings



25,000

Total Students



64.0

Average Overall Score



63.8

Average Mathematics Score



75.1%

Average Attendance

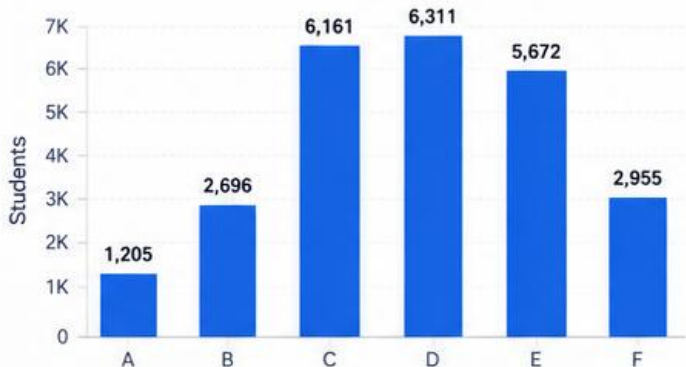


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Most Common Grade



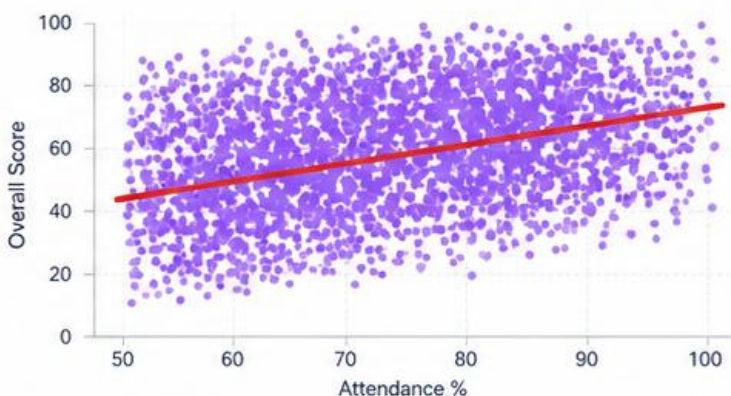
Final Grade Distribution



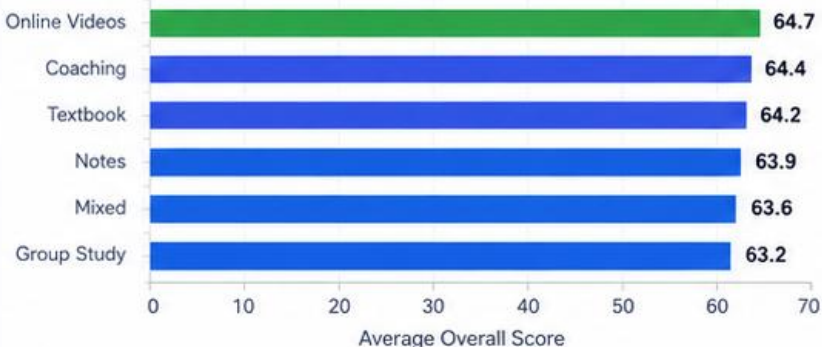
Average Score by Subject



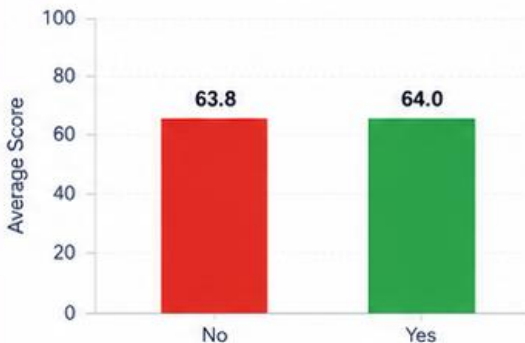
Attendance vs Overall Score



Average Score by Study Method



Internet Access



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Data is not just numbers, it's the story that drives better decisions.

– DIRISA



Executive Insights

- ✓ Study hours strongly influence performance.
- ✓ Mathematics requires attention.
- ✓ Attendance has a positive relationship with overall performance.
- ✓ Internet access supports performance.
- ✓ Grade D is the most common outcome.



Data is based on student records collected for academic performance analysis.

AN INITIATIVE OF



MANAGED BY



THANK YOU

NICIS

NATIONAL INTEGRATED
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